

Track 23: ROS2

June 2026

Track 23: ROS2 (Robot Operating System 2)

MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Linux → ROS2 Concepts → Nodes → Topics → Services → Navigation → Simulation → Real Robot

Related: [Track 19 — Robotics & IoT](#) | [Track 24 — Drone Programming](#)

Track Overview

Who this is for: Robotics engineers, ME/ECE students, and developers building autonomous robot software.

Prerequisites: Python or C++; Linux basics; [Arduino](#) or [Embedded Systems](#) helpful.

Tools: ROS2 Humble/Iron, Gazebo/Ignition, RViz2, TurtleBot3 (sim or hardware), Ubuntu 22.04, colcon.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	ROS2 awareness	Publisher/subscriber nodes
1 Month	80–100	Junior ROS developer	Simulated robot control
3 Months	250–300	Internship-ready	Navigation stack capstone
6 Months	500–600	Robotics engineer entry	Autonomous mobile robot
9 Months	750–900	Advanced robotics	Multi-robot coordination system

2-Week Crash Course

Days	Topics	Deliverable
D1–2	ROS2 architecture, workspace, colcon build	Working ROS2 workspace
D3–4	Nodes, topics, publishers/subscribers (Python)	Talker/listener nodes
D5–6	Services, parameters, launch files	Parameterized launch system
D7–8	RViz2, TF2 transforms	Visualized robot model
D9–10	Gazebo simulation intro	Simulated robot moving in Gazebo

1-Month Foundation

Week	Topics	Project
W1	ROS2 CLI, rclpy, package structure	Custom message package
W2	URDF, robot description, RViz2	Custom robot model
W3	Gazebo plugins, sensor simulation	Simulated LIDAR robot
W4	Basic navigation (Nav2 intro)	Point-to-point navigation in sim

3-Month Job Ready

Month	Topics	Projects
M1	ROS2 core, Python/C++ nodes, launch	Package suite (3 nodes)
M2	SLAM (Cartographer/SLAM Toolbox), mapping	Generated map of environment
M3	Nav2 stack, costmaps, capstone	Autonomous navigation capstone

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Manipulation intro (MoveIt2 overview)	Pick-and-place sim
M5	Real robot integration (TurtleBot or custom)	Hardware deployment
M6	Capstone	Autonomous mobile robot (sim + hardware)

9-Month Advanced

Capstones: Multi-robot fleet coordination + manipulation-capable robot system | **Level:** L5

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	ROS2 concepts, CLI, architecture
Lab Completion	40%	Nodes run; sim launches cleanly
Project	30%	Navigation accuracy, code structure
Portfolio	10%	GitHub repos, bag files, demo videos

Appendix: Mentor Notes

Use Ubuntu natively or VM with GPU passthrough for Gazebo. TurtleBot3 simulation sufficient for 1–3 month tiers; hardware optional for 6-month+.